ASSIGNMENT 3 DV LAB - POWER BI TASKS



SYMBIOSIS INSTITUTE OF TECHNOLOGY

UNDER THE GUIDANCE OF

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BLOG LINK -

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Assignment Practical No. 3 — Step-wise Task Log

Dataset: Adventure Works. Each task below includes a brief two-step explanation and a placeholder box to paste the corresponding screenshot.

A. Built-in Aggregations

Task 1: SUM() on ProductPrice

1. Drag ProductPrice into the Values/Card area and set its aggregation to Sum.

2. This displays the total of all product prices in the dataset.

Task 2: AVERAGE() on ProductPrice

1. Set the aggregation type of the ProductPrice field to Average.

2. This shows the mean price across all products.

Task 3: COUNT() on OrderNumber (Total Orders)

1. Apply Count aggregation on the OrderNumber field and place it on a Card visual.

2. This gives the total number of orders placed.

Task 4: DISTINCTCOUNT() on CustomerKey (Total Customers)

1. Apply Distinct Count aggregation on the CustomerKey field.

2. This counts unique customers, avoiding duplicate counting of repeat buyers.

Task 5: MIN() on ProductCost

1. Set the aggregation of the ProductCost field to Minimum.

2. This shows the lowest cost among all products.

Task 6: MAX() on ProductCost

1. Set the aggregation of the ProductCost field to Maximum.

2. This shows the highest cost among all products.

Task 7: SUM() on OrderQuantity

1. Apply Sum aggregation on the OrderQuantity field using a Card or Table visual.

2. This gives the total number of units ordered.

Task 8: SUM() on Total Products

1. Apply Sum aggregation on the Total Products field.

2. This gives the overall total/count of products across all records.

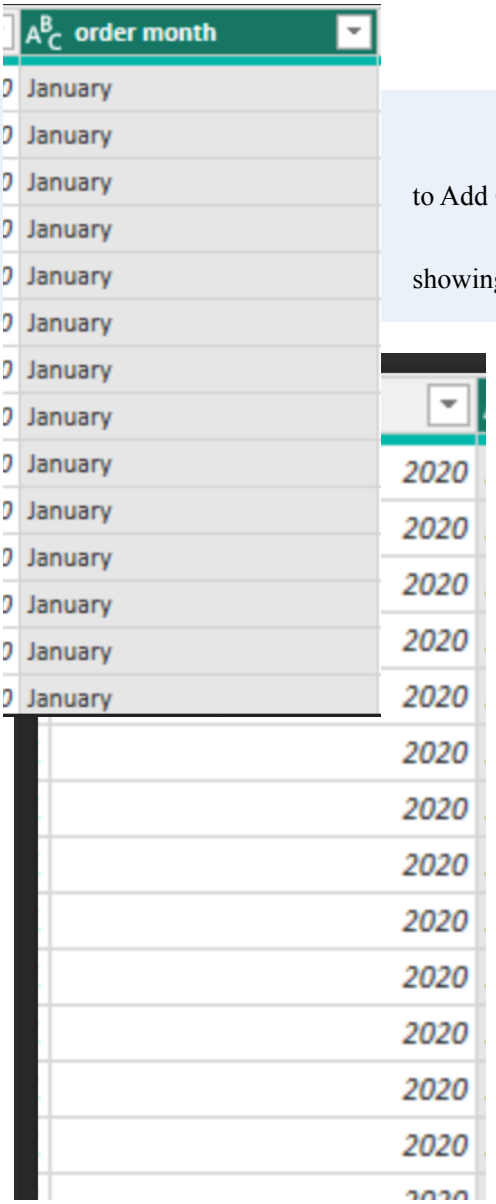


B. Calculated Columns (Using Power Query)

Task 9: Extract Order Year

1. Select the Date column, then go to Add Column > Date > Year > Year.

2. This creates a new column showing only the order year.



Task 10: Extract Month Name

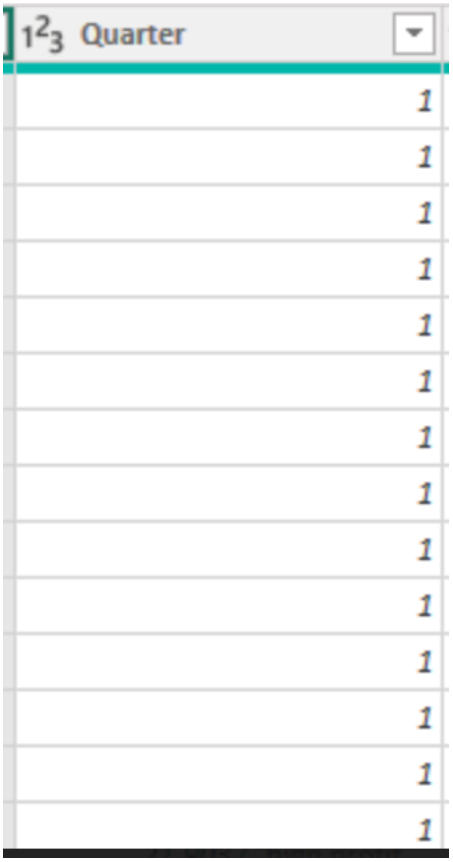
1. Select the Date column, then go to Add Column > Date > Month > Name of Month.

2. This creates a column showing the month name, e.g. January.

Task 11: Extract Quarter

1. Select the Date column, then go to Add Column > Date > Quarter > Quarter of Year.

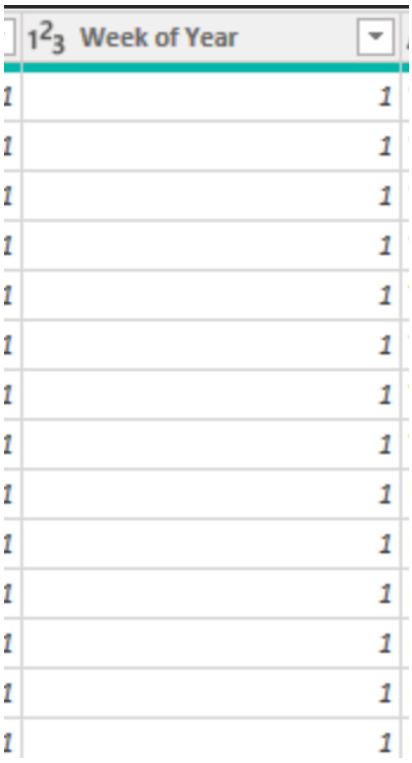
2. This creates a column showing which quarter the order falls in.



Task 12: Extract Week Number

1. Select the Date column, then go to Add Column > Date > Week > Week of Year.

2. This creates a column showing the week number within the year.



1²3 Week of Year	
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

Task 13: Extract Day Name

1. Select the Date column, then go to Add Column > Date > Day > Name of Day.

2. This creates a column showing the day name, e.g. Monday.

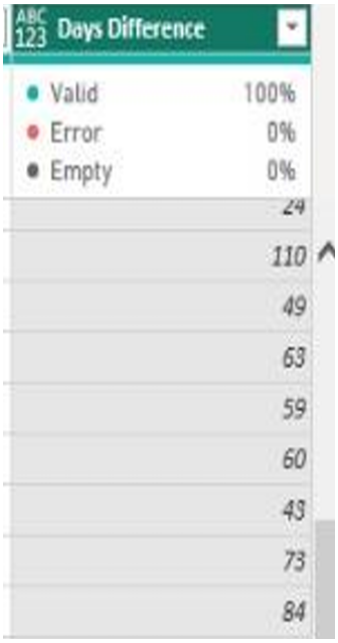


ABC Day Name
Wednesday
Wednesday
Wednesday
Wednesday
Thursday
Thursday
Thursday
Thursday
Friday
Friday
Friday
Friday
Friday
Friday

Task 14: Calculate Days Difference (OrderDate − StockDate)

1. Select both date columns, then go to Add Column > Date > Subtract Days.

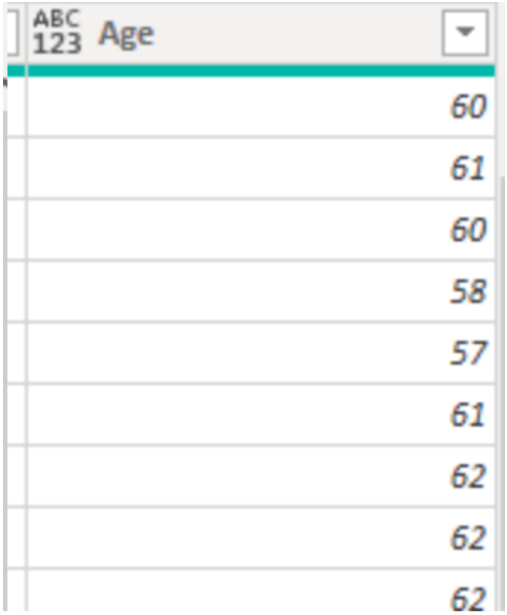
2. This creates a column showing the number of days between the two dates.



Task 15: Calculate Customer Age (if Birth Date exists)

1. Select the Birth Date column, then go to Add Column > Date > Age.

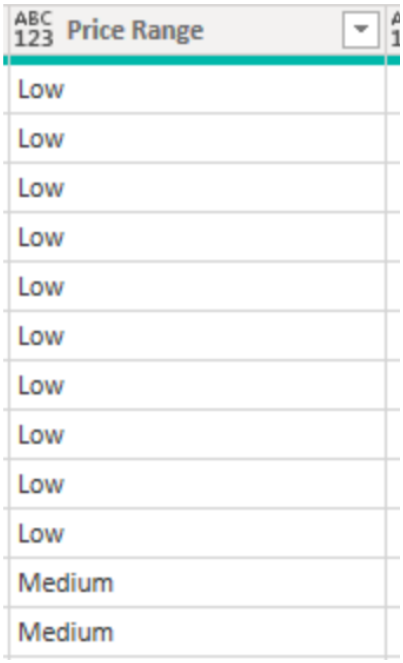
2. This calculates each customer's current age from their birth date.



Task 16: Create Price Range (Low/Medium/High)

1. Go to Add Column > Conditional Column and set rules based on ProductPrice ranges.

2. This labels each row as Low, Medium, or High price.

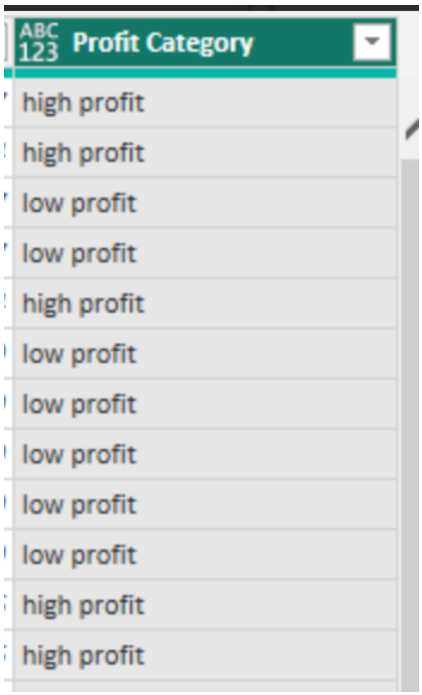


Price Range
Low
Low
Low
Low
Low
Low
Low
Low
Low
Low
Medium
Medium

Task 17: Create Profit Category (Loss/Low/High Profit)

1. Go to Add Column > Conditional Column and set rules based on profit value ranges.

2. This labels each row as Loss, Low Profit, or High Profit.

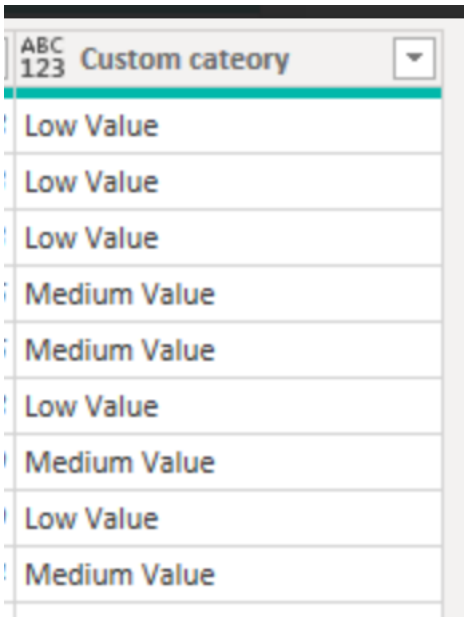


Profit Category
high profit
high profit
low profit
low profit
high profit
low profit
low profit
low profit
low profit
low profit
high profit
high profit

Task 18: Categorize Customers based on Sales

1. Go to Add Column > Conditional Column using each customer's total sales value.

2. This assigns a category label based on defined sales thresholds.

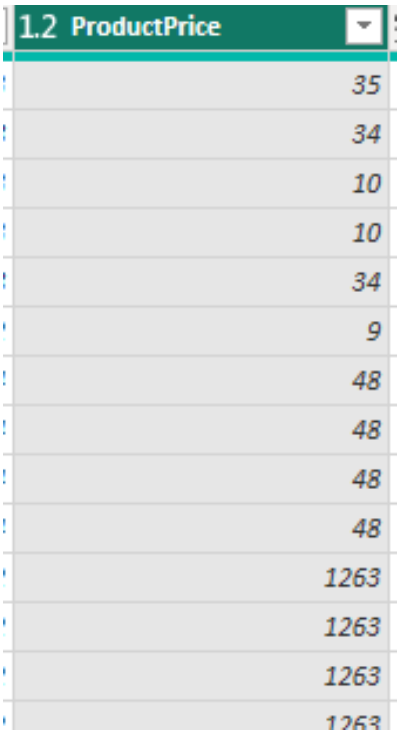


ABC 123 Custom cateory
Low Value
Low Value
Low Value
Medium Value
Medium Value
Low Value
Medium Value
Low Value
Medium Value

Task 19: Round Price Amount

1. Select the Price column, then go to Transform > Rounding > Round.

2. This rounds the price values to a specified number of decimals.



1.2 ProductPrice
35
34
10
10
34
9
48
48
48
48
1263
1263
1263
1263

Task 20: Round Cost Amount (2 decimals)

1. Select the Cost column, go to Transform > Rounding > Round and set decimal places to 2.

2. This standardizes cost values to two decimal places.

.2 ProductCost
13.09
12.03
3.4
3.4
12.03
5.71
31.72
31.72
31.72
31.72

Task 21: Convert Product Name to UPPERCASE

1. Select the Product Name column, then go to Transform > Format > UPPERCASE.

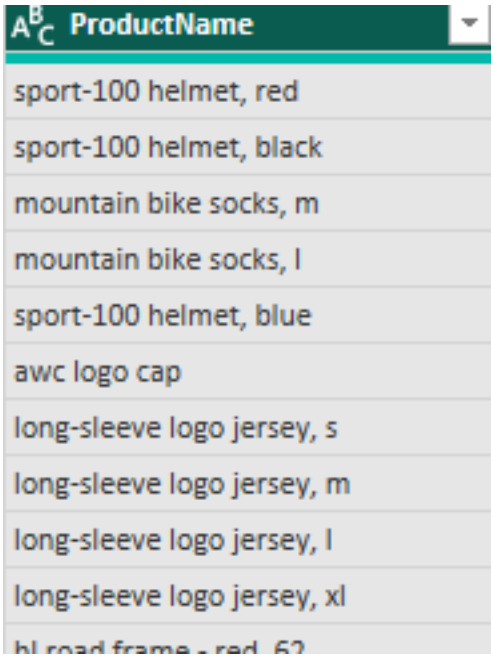
2. This converts all text in the column to capital letters.

ABC ProductName
SPORT-100 HELMET, RED
SPORT-100 HELMET, BLACK
MOUNTAIN BIKE SOCKS, M
MOUNTAIN BIKE SOCKS, L
SPORT-100 HELMET, BLUE
AWC LOGO CAP
LONG-SLEEVE LOGO JERSEY, S
LONG-SLEEVE LOGO JERSEY, M
LONG-SLEEVE LOGO JERSEY, L

Task 22: Convert Product Name to lowercase

1. Select the Product Name column, then go to Transform > Format > lowercase.

2. This converts all text in the column to small letters.

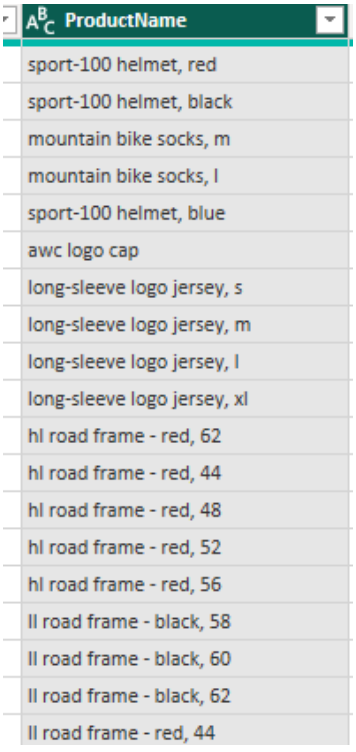


ProductName
sport-100 helmet, red
sport-100 helmet, black
mountain bike socks, m
mountain bike socks, l
sport-100 helmet, blue
awc logo cap
long-sleeve logo jersey, s
long-sleeve logo jersey, m
long-sleeve logo jersey, l
long-sleeve logo jersey, xl
hl road frame - red, 62

Task 23: Trim Extra Spaces

1. Select the Product Name column, then go to Transform > Format > Trim.

2. This removes leading and trailing spaces from the text.



ProductName
sport-100 helmet, red
sport-100 helmet, black
mountain bike socks, m
mountain bike socks, l
sport-100 helmet, blue
awc logo cap
long-sleeve logo jersey, s
long-sleeve logo jersey, m
long-sleeve logo jersey, l
long-sleeve logo jersey, xl
hl road frame - red, 62
hl road frame - red, 44
hl road frame - red, 48
hl road frame - red, 52
hl road frame - red, 56
ll road frame - black, 58
ll road frame - black, 60
ll road frame - black, 62
ll road frame - red, 44

Task 24: Clean Product Name

1. Select the Product Name column, then go to Transform > Format > Clean.

2. This removes non-printable characters from the text.

ProductName
sport-100 helmet, red
sport-100 helmet, black
mountain bike socks, m
mountain bike socks, l
sport-100 helmet, blue
awc logo cap
long-sleeve logo jersey, s
long-sleeve logo jersey, m
long-sleeve logo jersey, l
long-sleeve logo jersey, xl
hl road frame - red, 62
hl road frame - red, 44
hl road frame - red, 48
hl road frame - red, 52

Task 25: Capitalize Each Word

1. Select the Product Name column, then go to Transform > Format > Capitalize Each Word.

2. This capitalizes the first letter of every word in the text.

ProductName
Sport-100 Helmet, Red
Sport-100 Helmet, Black
Mountain Bike Socks, M
Mountain Bike Socks, L
Sport-100 Helmet, Blue
Awc Logo Cap
Long-Sleeve Logo Jersey, S
Long-Sleeve Logo Jersey, M
Long-Sleeve Logo Jersey, L
Long-Sleeve Logo Jersey, XL
Hl Road Frame - Red, 62

Task 26: Split Product Name by "-"

1. Select the Product Name column, then go to Split Column > By Delimiter and enter "-".

2. This splits the text into separate columns at each hyphen.

ProductSKU.1	ProductSKU.2	ProductSKU.3
HL	U509	R
HL	U509	*null*
SO	B909	M
SO	B909	L
HL	U509	B
CA	1098	*null*
LJ	0192	S
LJ	0192	M
LJ	0192	L
LJ	0192	X

Task 27: Extract First 8 Characters

1. Select the Product Name column, go to Add Column > Extract > First Characters and enter 8.

2. This creates a new column containing only the first 8 characters.

ProductName
Sport-10
Sport-10
Mountain
Mountain
Sport-10
Awc Logo
Long-Sle
Long-Sle
Long-Sle
Long-Sle
Hl Road
Hl Road
Hl Road
Hl Road
Hl Road

Task 28: Extract Last 5 Characters

1. Select the Product Name column, go to Add Column > Extract > Last

Characters and enter 5.

2. This creates a new column containing only the last 5 characters.



Task 29: Merge Product Name and Color

1. Select both columns, then go to Add Column > Merge Columns and choose a separator.

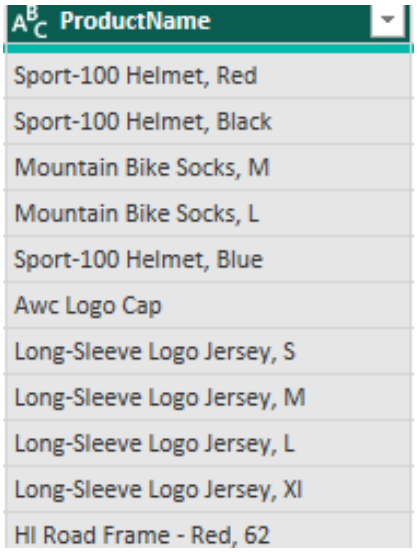
2. This combines the two columns into a single descriptive column.



Task 30: Duplicate Product Name Column

1. Right-click the Product Name column and select Duplicate Column.

2. This creates an exact copy of the column for further transformation.



ProductName
Sport-100 Helmet, Red
Sport-100 Helmet, Black
Mountain Bike Socks, M
Mountain Bike Socks, L
Sport-100 Helmet, Blue
Awc Logo Cap
Long-Sleeve Logo Jersey, S
Long-Sleeve Logo Jersey, M
Long-Sleeve Logo Jersey, L
Long-Sleeve Logo Jersey, XI
HI Road Frame - Red, 62

Task 31: Replace Missing Values

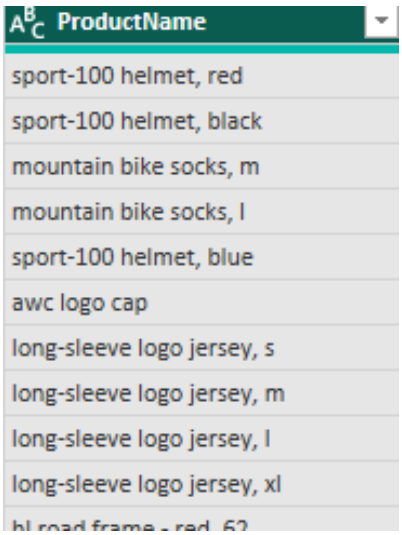
1. Select the column, right-click and choose Replace Values, entering null and a default value.

2. This fills in missing or blank entries with a specified value.

Task 32: Replace Bike with Bicycle

1. Select the Product Name column, right-click and choose Replace Values.

2. This replaces every occurrence of "Bike" with "Bicycle".



ProductName
sport-100 helmet, red
sport-100 helmet, black
mountain bike socks, m
mountain bike socks, l
sport-100 helmet, blue
awc logo cap
long-sleeve logo jersey, s
long-sleeve logo jersey, m
long-sleeve logo jersey, l
long-sleeve logo jersey, xl
hl road frame - red, 62

Task 33: Change Sales Data Type to Fixed Decimal

1. Select the Sales column and set the Data Type dropdown to Fixed Decimal Number.

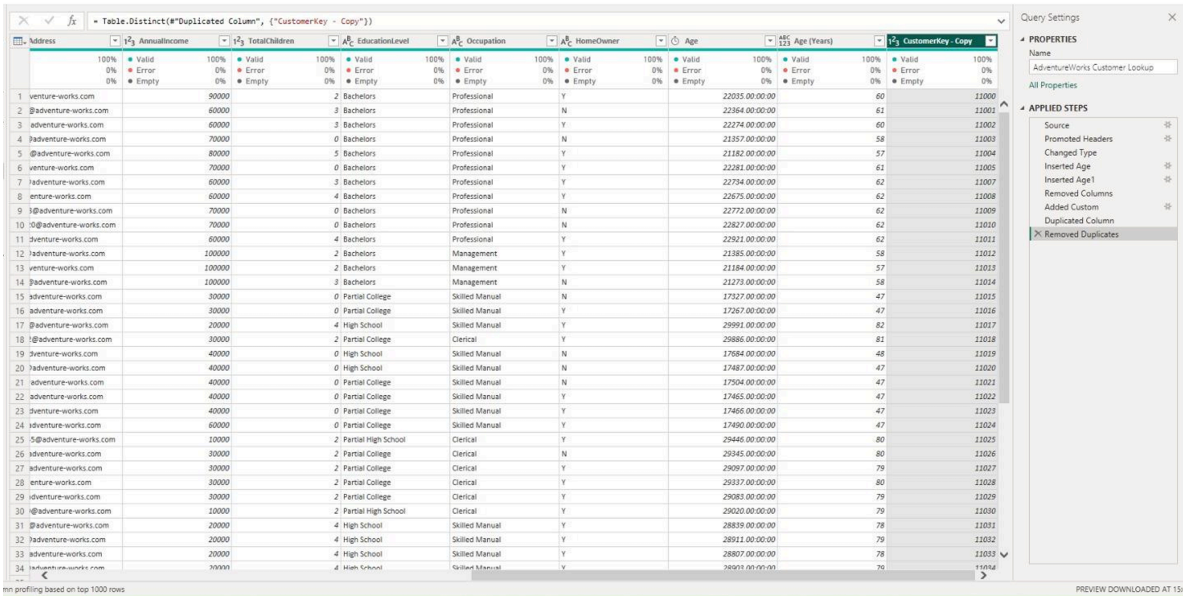
2. This ensures sales values are stored in a consistent decimal format.



Task 34: Remove Duplicate Customers

1. Select the Customer column(s), right-click and choose Remove Duplicates.

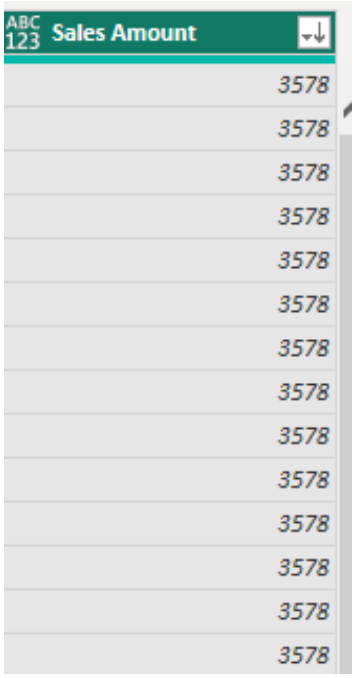
2. This keeps only unique customer records in the table.



Task 35: Sort Products by Sales Amount

1. Select the Sales column and click Sort Ascending or Sort Descending.

2. This arranges the products according to their sales value.

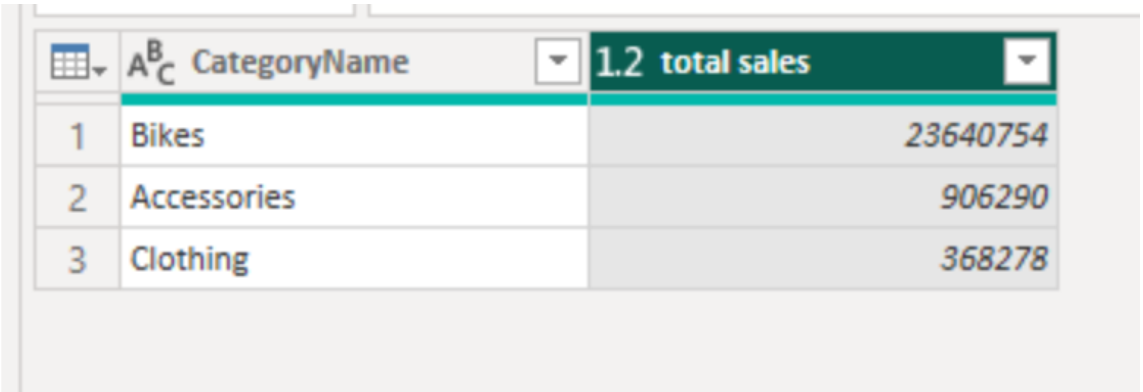


Sales Amount
3578
3578
3578
3578
3578
3578
3578
3578
3578
3578
3578
3578
3578

Task 36: Group by Product Category and Sum Sales

1. Select the Product Category column, go to Transform > Group By and set aggregation to Sum of Sales.

2. This creates a summary table of total sales per category.



	CategoryName	total sales
1	Bikes	23640754
2	Accessories	906290
3	Clothing	368278

Task 37: Group by Customer and Average Sales

1. Select the Customer column, go to Transform > Group By and set aggregation to Average of Sales.

2. This creates a summary table of average sales per customer.

CustomerKey	avg sales
12885	1142.4
11818	1147.4
11872	1141.4
12717	1879
20215	2639.5
11671	1435.25
11809	1433.25
20227	1931
11834	1887
20156	1379.666667
11773	1439.25
20239	1167.2
20216	1340.25
15631	1030.6
19977	1161.2
12805	898.1428571

Task 38: Create Index Column

1. Go to Add Column > Index Column > From 1.

2. This adds a sequential row-number column to the table.

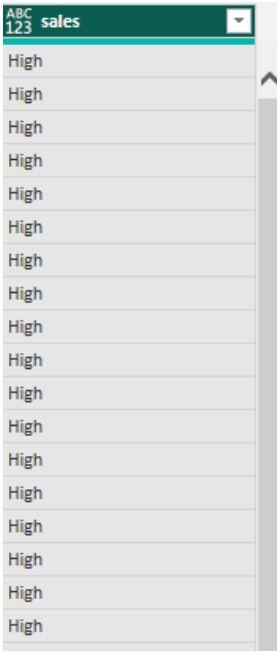
Index
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14

C. Additional Tasks

Task 39: Create Sales Band (₹0–₹500 Low, ₹501–₹2000 Medium, Above ₹2000 High)

1. Go to Add Column > Conditional Column and set three rules based on the Sales value range.

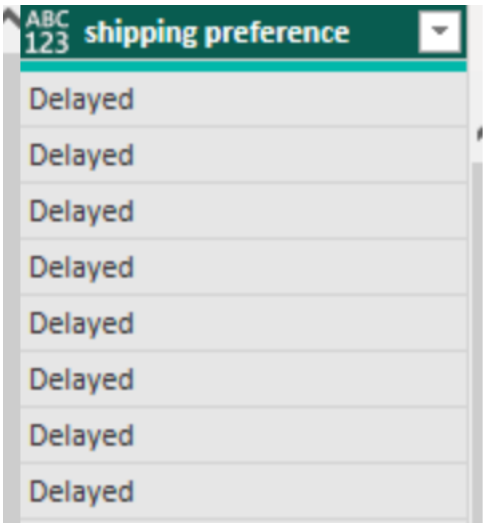
2. This labels each row as Low, Medium, or High sales band.



Task 40: Create Shipping Performance (0–3 Days Fast, 4–7 Days Normal, 7+ Days Delayed)

1. Go to Add Column > Conditional Column using the days-difference between order and ship date.

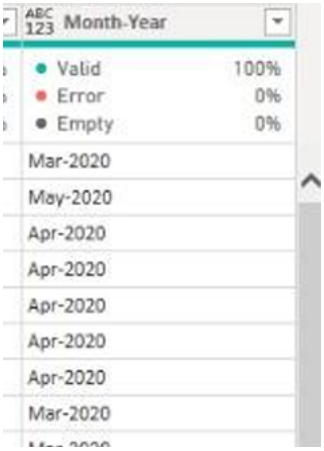
2. This labels each row as Fast, Normal, or Delayed shipping.



Task 41: Extract Month-Year from Order Date (e.g., Jan-2014)

1. Use Add Column > Custom Column with a formula combining Month name and Year.

2. This creates a single column displaying the order's month and year together.



Task 42: Split Customer Full Name into First Name and Last Name

1. Select the Full Name column, go to Split Column > By Delimiter using a space as separator.

2. This creates two separate columns for First Name and Last Name.

Task 43: Merge Product Category and Subcategory into a Single Column

1. Select both columns, go to Add Column > Merge Columns and choose a separator such as "-".

2. This combines category and subcategory into one descriptive column.

Task 44: Create Discount Percentage Band (No/Low/Medium/High Discount)

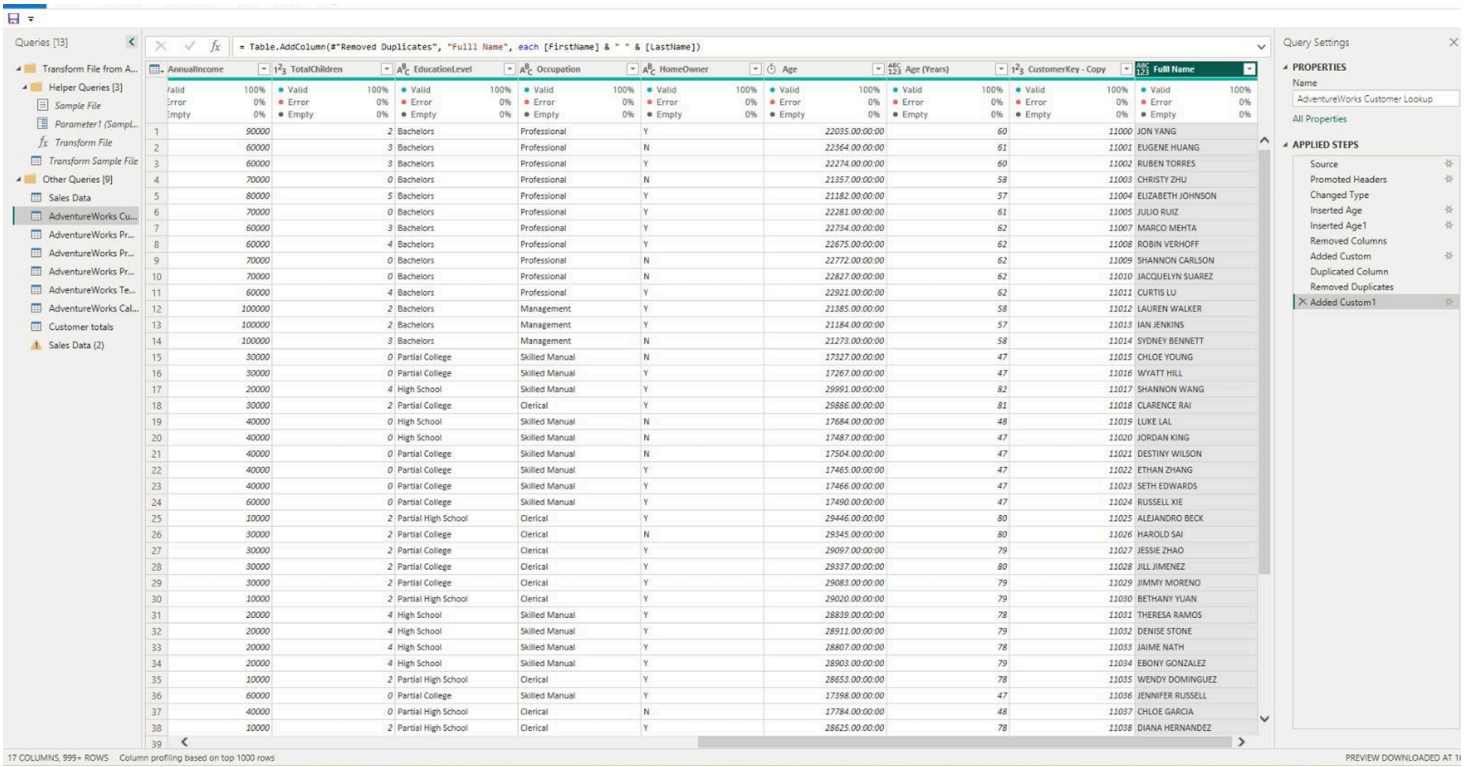
1. Go to Add Column > Conditional Column and set rules based on Discount Percentage ranges.

2. This labels each row as No, Low, Medium, or High Discount.

Task 45: Group by Region and Calculate Total Sales, Average Sales, Total Quantity

1. Select the Region column, go to Transform > Group By and add Sum of Sales, Average of Sales, and Sum of Quantity.

2. This produces a summary table of key metrics for each region.



	AnnualIncome	TotalChildren	EducationLevel	Occupation	HomeOwner	Age	Age (Years)	CustomerKey - Copy	Full Name
1	90000	2	Bachelors	Professional	Y	22035.00:00:00	60	11000	JON YANG
2	60000	3	Bachelors	Professional	N	22364.00:00:00	61	11001	EUGENE HUANG
3	60000	3	Bachelors	Professional	Y	22274.00:00:00	60	11002	RUBEN TORRES
4	70000	0	Bachelors	Professional	N	21357.00:00:00	58	11003	CHRISTY ZHU
5	80000	5	Bachelors	Professional	Y	21182.00:00:00	57	11004	ELIZABETH JOHNSON
6	70000	0	Bachelors	Professional	Y	22281.00:00:00	61	11005	JULIO RUIZ
7	60000	3	Bachelors	Professional	Y	22734.00:00:00	62	11007	MARCO MEHTA
8	60000	4	Bachelors	Professional	Y	22675.00:00:00	62	11008	ROBIN VERHOFF
9	70000	0	Bachelors	Professional	N	22772.00:00:00	62	11009	SHANNON CARLSON
10	70000	0	Bachelors	Professional	N	22827.00:00:00	62	11010	JACQUELYN SUAREZ
11	60000	4	Bachelors	Professional	Y	22921.00:00:00	62	11011	CURTIS LU
12	100000	2	Bachelors	Management	Y	21385.00:00:00	58	11012	LAUREN WALKER
13	100000	2	Bachelors	Management	Y	21184.00:00:00	57	11013	IAN JENKINS
14	100000	3	Bachelors	Management	N	21273.00:00:00	58	11014	SYDNEY BENNETT
15	30000	0	Partial College	Skilled Manual	N	17327.00:00:00	47	11015	CHLOE YOUNG
16	30000	0	Partial College	Skilled Manual	Y	17267.00:00:00	47	11016	WYATT HILL
17	20000	4	High School	Skilled Manual	Y	29991.00:00:00	82	11017	SHANNON WANG
18	30000	2	Partial College	Clerical	Y	29886.00:00:00	81	11018	CLARENCE RAI
19	40000	0	High School	Skilled Manual	N	17684.00:00:00	48	11019	LUKE LAL
20	40000	0	High School	Skilled Manual	N	17487.00:00:00	47	11020	JORDAN KING
21	40000	0	Partial College	Skilled Manual	N	17504.00:00:00	47	11021	DESTINY WILSON
22	40000	0	Partial College	Skilled Manual	Y	17465.00:00:00	47	11022	ETHAN ZHANG
23	40000	0	Partial College	Skilled Manual	Y	17466.00:00:00	47	11023	SETH EDWARDS
24	60000	0	Partial College	Skilled Manual	Y	17490.00:00:00	47	11024	RUSSELL XIE
25	10000	2	Partial High School	Clerical	Y	29446.00:00:00	80	11025	ALEJANDRO BECK
26	30000	2	Partial College	Clerical	N	29345.00:00:00	80	11026	HAROLD SAI
27	30000	2	Partial College	Clerical	Y	29097.00:00:00	79	11027	JESSIE ZHAO
28	30000	2	Partial College	Clerical	Y	29337.00:00:00	80	11028	JILL JIMENEZ
29	30000	2	Partial College	Clerical	Y	29083.00:00:00	79	11029	JIMMY MORENO
30	10000	2	Partial High School	Clerical	Y	29020.00:00:00	79	11030	BETHANY YUAN
31	20000	4	High School	Skilled Manual	Y	28839.00:00:00	78	11031	THERESA RAMOS
32	20000	4	High School	Skilled Manual	Y	28911.00:00:00	79	11032	DENISE STONE
33	20000	4	High School	Skilled Manual	Y	28807.00:00:00	78	11033	JAIME NATH
34	20000	4	High School	Skilled Manual	Y	28903.00:00:00	79	11034	EBONY GONZALEZ
35	10000	2	Partial High School	Clerical	Y	28653.00:00:00	78	11035	WENDY DOMINGUEZ
36	60000	0	Partial College	Skilled Manual	Y	17398.00:00:00	47	11036	JENNIFER RUSSELL
37	40000	0	Partial High School	Clerical	N	17784.00:00:00	48	11037	CHLOE GARCIA
38	10000	2	Partial High School	Clerical	Y	28625.00:00:00	78	11038	DIANA HERNANDEZ
39									